COURSE CODE	COURSE TITLE	L	T	P	TW & SL	TH	C
UCS3504	SOFTWARE ENGINEERING	45	0	0	45	90	3

OBJECTIVES

- To understand the phases in a software project, estimate cost and effort.
- To understand fundamental concepts of requirements engineering and Analysis Modeling.
- To understand the various software design methodologies
- To learn various testing techniques and maintenance measures.
- To understand agile development and ML Devops

UNIT I	SOFTWARE PROCESS AND PLANNING	9

Introduction to Software Engineering; Objectives, Principles and Practices; The Software Development Life Cycle : Pre-development phases of the SDLC — Development specific phases of the SDLC — Post-development phases of the SDLC; Methodologies Paradigm and Practices : Process methodologies — Development paradigms — Development practices; Project Planning Process; Software Project Estimation: Decomposition techniques — Empirical estimation models —Project scheduling; Risk Management; Project delivery management.

UNIT II	REQUIREMENTS ANALYSIS AND SPECIFICATION	9

Software Requirements: Functional and non-functional — Security requirements — User requirements — System requirements — Software requirements document; Requirement Engineering Process: Feasibility studies — Requirements elicitation and analysis — Requirements validation — Requirements Management; Classical Analysis: Structured system analysis; Requirement modelling tools.

UNIT III	SOFTWARE DESIGN	9

Design Concepts: Design process — Design concepts — Modularity, Coupling and cohesion — Design model — Modeling principles; Structured Design; Architectural Design: Architectural styles; Architecture for Network based Applications — Decentralized Architectures.

UNIT IV	SOFTWARE TESTING	9

Software Testing Fundamentals; Internal and External Views of Testing: White box testing — Basis path testing — Control structure testing— Black box testing — Unit testing — Integration testing — Regression testing — Validation testing — System testing — Security testing; Testing Tool; Debugging; Software Implementation: Coding Practices and Principles; Maintenance: Types

UNIT V	AGILE DEVELOPMENT AND DEVOPS	9

Agile Development: Agile Teams — Team and Scrum — Branches — Pull Requests — Reviews — Integration — Agile Iterations — Reporting and fixing bugs; ML Dev/Ops: From development to deployment — Three-Tier — Responsiveness, Service level objectives, and Apdex — Releases and feature flags — Monitoring and finding bottlenecks — Improving rendering and database performance with caching; Security: Defending customer data in application.

TOTAL PERIODS	**90**

COURSE OUTCOMES On successful completion of this course, students will be able to
CO1 Choose an appropriate process model and estimate project cost and effort required by applying software engineering principles (K5)
CO2 Identify and analyze the requirements and construct their models (K5)
CO3 Apply systematic procedure for software design (K3)
CO4 Compare and contrast the various testing and maintenance activities (K2)
CO5 Make use of agile development and DeVops (K3)
CO6 Identify unethical issues and apply ethical practices for a given case study (K3)

TEXT BOOKS
1. Roger S. Pressman, Bruce R. Maxim, *Software Engineering: A Practitioner's Approach*, McGraw-Hill International Edition, 9th Edition, 2019. (All units)
2. Armando Fox and David Patterson, Engineering Software as a Service: An Agile Approach Using Cloud Computing", Strawberry Canyon LLC, Second Beta Edition, 2021. (Unit 5)

REFERENCE BOOKS
1. Ian Sommerville, "Software Engineering", Pearson Education Asia,Tenth Edition, 2015.
2. Stephen R Schach, "Software Engineering", Tata McGraw-Hill Publishing Company Limited, 2007.
3. Brian Albee, Hands-On Software Engineering with Python, Packt Publishing, 2018.
4. Kelkar S A,"Software Engineering", Prentice Hall of India, 2007.

CO-PO-PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2										3
CO2	2	3		2									2
CO3		2	3		3								3
CO4			3	3									2
CO5					3				3				3
CO6						2	3	3					2